

Full MVP Specification — Best Invest Properties

Date: 25 September 2026

Target platform: Bubble.io

Integrations: Make, OpenAI API, transactional email provider

This specification is based on the approved visual prototype, which defines the interface and visible behaviour. Numbers in the prototype are test data.

1. Executive summary

Best Invest Properties is a marketplace / controlled introduction platform for investment property in Spain and Cyprus. The platform lets:

- an investor find a specific available unit, compare financial metrics, see the score and request an introduction;
- a developer get verified, submit a project and keep price and availability up to date;
- the Best Invest team review all public data, the score, the AI narrative and every contact disclosure.

The MVP is not a financial adviser, a transaction platform or a full CRM. Its value is a controlled catalogue with transparent metrics, reviewed written analysis and a managed lead/introduction flow.

2. Goals and success metrics

Product goals

1. Give the investor a clear path from criteria to an enquiry on a specific unit.
2. Give the verified developer a managed self-service submission flow.
3. Keep editorial/approval control with Best Invest.
4. Make every score, yield and narrative reproducible and auditable.
5. Launch Spain/Cyprus without hard-coded country logic, so adding a new market does not require rebuilding the database.

Launch KPIs

KPI	Initial target
Search → property detail CTR	measure baseline; target after 30 days
Property detail → enquiry conversion	≥ 3% qualified traffic
Registration completion	≥ 55% of those who started the form
Developer application completion	≥ 40%

KPI	Initial target
Project submission without support intervention	≥ 70%
Admin introduction decision within SLA	≥ 90%
Duplicate enquiries/automations	0 critical duplicates
Published records with complete traceability	100%

3. Roles

Role	Main permissions
Anonymous visitor	landing, search, public unit detail, investment analysis and score
Investor	save, saved search, calculator scenarios, enquiry, dashboard, settings/data rights
Developer applicant	application status, requested information, draft preparation by rule
Verified developer member	own company/projects/units, change requests, masked leads
Admin	developer verification, project review and publication, introduction decisions, score model, users and automations

One User may have several roles, but the MVP UI shows one active workspace at a time.

In the MVP there is one admin type, `admin`. The "four eyes" rule does not apply; instead, confirmation is mandatory for every important action, with a full history in the audit log. The admin area has the navigation Dashboard · Approvals · Investors · Investment Scores.

Architecture requirement: the access model is built so that `support`, `reviewer` and `senior_admin` roles can be added later without rework. So permissions are checked through `admin_permission_keys` in backend workflows, not by checking "role = admin".

4. Architecture decisions (ADR)

ID	Decision	Consequence
AD R-01	Investor browses Unit	search/detail/enquiry tied to concrete availability/price
AD R-02	Project → Unit Type → Unit	repeated floor plans without duplication; individual price/status
AD R-03	Single villa = one-unit Project	one model and one workflow
AD R-04	One User, multiple roles	single auth store; role-based routing
AD R-05	Application creates User/Company immediately	applicant can see status and add data

ID	Decision	Consequence
AD R-06	Enquiry is aggregate root	one canonical lifecycle; role-specific labels are presentation only
AD R-07	Score/financials deterministic, AI narrative only	figures reproducible; AI failure does not block the core product
AD R-07a	Financial input estimates are produced by Make from collected comparable listings, not by AI; they are not published without admin approval; all derived values and the score are calculated by the deterministic Bubble service	the principle "OpenAI does not calculate money" applies without exception
AD R-08	Bubble source of truth, Make orchestrator	simpler audit, retry, privacy and recovery

5. Technical architecture

flowchart LR

```

U[Users] --> B[Bubble responsive web app]
B --> DB[(Bubble database + private files)]
B --> BW[Bubble backend workflows]
BW --> M[Make scenarios]
M --> O[OpenAI Responses API]
BW --> E[Email via SendGrid]
M --> BW
BW --> DB
A[Admins] --> B

```

Component responsibilities

- Bubble pages: UI, route guards, forms, client calculations for preview only.
- Bubble backend workflows: authorisation, validation, deterministic calculations, status transitions, final writes.
- Bubble database: authoritative state, versions, audit, consent, integration jobs.
- Make: collecting rental listings from portals and generating the AI narrative via OpenAI; retry of external calls.
- OpenAI: reviewed narrative only.
- Email: all emails are sent by Bubble (backend workflows, own SendGrid key and domain with SPF/DKIM) — see FR-12.

6. Information architecture and screens

The "Prototype" column shows the screen number in the approved prototype.  means a minimal screen that is being added to the prototype (contents in §6.4).

Public / investor

ID	Screen	Prototype	MVP status
P01	Landing / Home	01	approved
P02	Investment Search & Results (Browse)	02	approved
P03	Property/Unit Detail	04	approved
P04	Investment Analysis	05	approved
P05	Financial Calculator	06	approved
P06	Compare Investments	07	approved
P07	Investor Registration	08	approved
P08	Login	15	approved
P09	Forgot Password / expired link	24	approved
P10	Investor Dashboard	09	approved
P11	Saved Searches	09 (panel)	panel on the P10 dashboard, no separate screen
P12	Enquiry Detail	—	minimal screen — §6.4
P13	Account Settings	25	approved
P14	Privacy, Terms	17, 18	layout approved; text prepared by the lawyer
P14 b	Cookie Policy, Investment/AI Disclaimer	—	minimal pages — §6.4
P15	How It Works / About / Contact	partly	"How it works" is an anchor section on P01; no separate About/Contact

Developer

ID	Screen	Prototype	MVP status
D0 1	For Developers	16	approved
D0 2	Developer Sign-up	19	approved
D0 3	Verification Status / Re-upload	26	approved
D0 4	Developer Portal	10	approved
D0 5	Company Profile	27	approved
D0 6	Add/Edit Draft Project	11	approved
D0 7	My Projects	20	approved

ID	Screen	Prototype	MVP status
D08	Project & Units	21	approved
D09	Changes Requested	10 (changes state)	implemented as a state of the D04 portal, not a separate screen
D10	Leads	22	approved
D11	Lead Detail / outcome	—	minimal screen — §6.4

Admin

ID	Screen	Prototype	MVP status
A01	Admin Login / role gate	31	approved
A02	Dashboard	12	approved
A03	Unified Approvals Queue	23	approved; four sections (see §6.1)
A04	Project Review	28	approved
A05	Developer Verification	29	approved
A06	AI / Analysis Review	28 (panel)	panel inside A04, no separate screen — §6.2
A07	Score Model Versions	13	approved; includes a compact versions table — FR-11b
A08	Enquiries / contact release	12 + 23	split: enquiries table on A02, decisions in the Introductions section of A03
A09	Users and Companies lookup	30	approved as "User Management" (investors + developer contacts)
A10	Automation Monitor	—	minimal block — §6.4
A11	Country & Cost Settings	—	minimal form — §6.4
A12	User Journeys (internal diagram)	14	reference screen for the team; not part of the client-facing product

The admin navigation has exactly four items: **Dashboard** • **Approvals** • **Investors** • **Investment Scores**. A04 and A05 open from the A03 queue and have a "← Approvals queue" back button; they have no menu items of their own.

6.1 Approvals queue structure (A03)

One queue with a type switch — five tabs: **All** • **Project submissions** • **Developer applications** • **Introductions** • **Listing changes**. Sections:

Section	Source	Row actions
New project submissions	Project submitted	"Open review" → A04
Developer applications	Developer Application submitted/under_review	"Open verification" → A05
Investor introductions	Enquiry awaiting a decision	Approve & connect · Hold · Decline
Listing change requests	Change Request for price/availability	Approve · Query · Reject

Each section shows an "N WAITING" counter that goes down after a decision. After a decision the row is replaced by a result block listing the emails sent (chips "✉ Investor notified", "✉ Developer notified") and a **Reopen** button (for Hold / Decline) or **Correct status** (for Connected) — see FR-11. "Queue clear" is a separate empty state for the whole screen.

Hold, Decline, Query and Reject require a written reason (the modal cannot be confirmed without it). Hold and Query additionally require a follow-up date and an owner — in the MVP the owner is filled in automatically. Approve for an introduction or a project submission needs no reason but shows a warning that contact disclosure cannot be undone.

6.2 Analysis panel in Project Review (A04)

The A04 screen contains an "Investment analysis" panel with two states (Awaiting review / Reviewed & approved) and these blocks:

1. **Metrics:** Investment score, Gross yield, Net yield.
2. **Proposed financial estimates** — rows with a provenance label: Proposed (proposed by the analyst), Assumption (platform default value), Calculated (calculated deterministically). These values are not produced by AI — see FR-03a.
3. **Category assessments** — five criteria with points and an explanation.
4. **Sources, assumptions & gaps** — tags SOURCE, DEVELOPER, ESTIMATE, GAP.
5. Note: "Analysis uses the agreed free/open sources. Information gaps are flagged for review or manual input".

Before approval the panel is labelled "Not yet visible to investors"; after — "Published ... Investors see this saved version".

6.3 Screen states

The list of states below is a mandatory part of the implementation scope.

Screen	States
P02 Browse	Results · Loading (skeletons) · Query failed (error with retry); separately Signed in / Signed out ; empty filter result
P03 Detail	Published · Enquiry sent · Analysis not ready · Withdrawn / 404
P04 Analysis	Full analysis · Narrative unavailable (deterministic fallback)
P05 Calculator	field validation; the result block is not calculated while there are errors
P06 Compare	empty (<2 properties) · 2-3 properties · limit when trying to add a 4th
P08 Login	Credentials · Two-factor · Locked ; login error state

Screen	States
P09 Forgot password	Request · Sent · Reset · Done
P10 Dashboard	Active investor · First day – nothing yet (empty state)
P13 Account settings	idle · Saving... · Saved
D04 Developer portal	verified · pending · changes · approved
D03 Verification status	pending · more info required · rejected ; document upload progress
D06 Add project	required-field errors; file upload progress and error
A02 Admin dashboard	Queue has work · Nothing to review ; decision result per card
A03 Approvals queue	Queue has work · Queue clear ; decision result per row
A04 Project review	Awaiting review · Reviewed & approved
A07 Score editor	idle · running (progress N of 83) · done · failed

Cross-cutting system states are also mandatory: 403 , maintenance, session expired, offline/network error, stale record/conflict, duplicate submit, unsaved changes guard. 404 is shown as the "Withdrawn" state on P03.

Toast and modals are global. One toast (auto-dismiss ~3.8 s, click to close) and one confirmation modal for the whole app. The modal has: a title, text, an optional note, an optional required reason field, and Cancel / confirm buttons. Confirmation tone: regular (blue) or destructive (dark red).

6.4 Minimal screens

Five screens are needed for the MVP **in minimal form** — one screen each, in the style of the approved design.

Screen	Minimal contents
A10 Automation Monitor	a small block in the admin area: list of stuck/failed jobs, type, entity, error text (redacted), Retry button. No large separate dashboard
A11 Country & Cost Settings	a form for editing taxes and costs per country. Mandatory: these values must not be edited directly in the database
P12 Investor Enquiry Detail	property/unit, enquiry date, current status (investor label), status history, contact disclosure state
D11 Developer Lead Detail	lead reference, unit/project, status, contacts permitted for display, a field for the lead outcome
P14b Cookie Policy + disclaimers	pages with the Privacy/Terms structure; text prepared by the lawyer. Cookie banner if non-essential analytics or marketing cookies are used

7. End-to-end flows

7.1 Investor discovery → enquiry

1. Visitor sets country, budget, bedrooms/type and minimum yield on Landing.
2. P02 opens with query parameters and live search constraints.
3. Visitor sorts/filters, saves to compare in session.
4. P03 shows one Unit with project context, approved media, availability, financials and score.
5. P04 explains the deterministic score; the AI narrative is labelled reviewed/AI-assisted.
6. P05 recalculates the scenario locally and optionally saves after login.
7. P06 compares 2-3 Units; each row has a canonical better direction.
8. "Request information/analysis" opens login/registration when anonymous.
9. Investor reviews contact-sharing consent and submits the Enquiry.
10. Confirmation shows the reference and `submitted` status.
11. Admin screens the enquiry; on approval, confirms the contact release.
12. Bubble sends both introduction emails; the Enquiry moves to `introduced` after both are sent.
13. Investor sees the history in P12; developer sees a role-safe status in D10/D11.

7.2 Developer application → verified portal

1. D02 creates the User, Developer Company and Application draft.
2. Applicant fills in company/contact/markets and uploads private documents.
3. Submit validates required fields/document matrix and records consents.
4. D03 shows `submitted/under_review` with the application reference.
5. Admin A05 reviews document by document.
6. `more_info_required` unlocks the relevant uploads/fields and shows the public reviewer message.
7. An approved application links the User to the Developer Company and activates the developer role.
8. A rejected application keeps the reason, support path and allowed resubmission policy.

7.3 Project submission → publication

1. Verified developer creates a Project draft.
2. Adds Unit Types, Units, media and private documents.
3. The readiness checklist is calculated from explicit submission rules.
4. Save draft is autosave/manual; the unsaved changes guard works.
5. Submit sets status `submitted`; the project cannot be edited during review.
6. Admin A04 sees full media/documents/financial inputs and a validation summary.
7. Request Changes creates a structured request; D09 shows fields/reason and the resubmit path.
8. Approve and Publish are separate actions.
9. Publish requires admin permission and a complete current score/analysis.
10. Published Units appear in P02; Project/Unit history remains reconstructable.

7.4 Price/availability update

1. Developer edits only the allowed fields on D08: **price** and **availability**.

2. The UI collects a Change Request batch with old/new values.
3. Submit freezes the items and shows pending per Unit ("Price change pending approval" in the unit row).
4. Admin A03 approves/rejects/queries each batch in the **Listing change requests** section.
5. Approval applies once, recalculates financials/score, marks AI stale and queues regeneration.
6. Investors who saved the unit receive the configured notifications.

Rules:

- **both** changes — price and availability — go through prior admin approval; there is no immediate availability change;
- availability values the developer can set: Available, Reserved, Sold;
- Withdrawn is not available to the developer — it is an admin action (P03 has the matching "Withdrawn / 404" state);
- all other fields of a published project are locked and shown in the **"LOCKED — ADMIN REVIEW REQUIRED"** panel: project name, location, completion, unit mix, specification, media. A request to change them is queued alongside the original submission;
- the D08 screen keeps a **Change history** with rows "date · what changed · state" (Awaiting approval, Approved).

8. Functional requirements

FR-01 Search and catalogue

- Search object: published and available Unit.
- Filters (left column, applied immediately, no Apply button):

Filter	Control	Values
Country	checkboxes	Cyprus, Spain
Property type	checkboxes	Apartment, Villa, House
Bedrooms	checkboxes	Studio, 1, 2, 3+
Strategy	checkboxes	Long-term rental, Short-term rental, Mixed with private use, Capital growth
Completion	checkboxes	Ready, <12 months, 12–24 months
Max price	slider	up to €400,000
Min gross yield	slider	from 6%
Min net yield	slider	from 4.5%

- Default sort: **net yield descending**.
- Sort options: **net yield, gross yield, investment score, price**.

- Debounce filter changes; update count and ranking. The match counter sits in the filter column header and shows `0` on an empty result.
- Rank number is the rank within the current filtered result set.
- **Pagination: 8 cards per page**, numbered pager with "Previous / Next" and the label "Showing X-Y of N properties". No infinite scroll.
- **"Top 5" rail** — a side block with the five best properties by the current sort; the label changes: `BY NET YIELD` / `BY GROSS YIELD` / `BY SCORE` / `BY PRICE`. On mobile it moves above the list. Hidden in the `loading` and `error` states.
- Empty state recommends widening named constraints.
- On mobile, filters collapse into a drawer with a selected-count badge.
- Compare is selected directly from result cards; limit is 3 properties, trying to add a fourth shows a toast.
- Anonymous compare persists in the browser session; account persistence optional in the MVP.
- Greece/Portugal never look live unless Country Config says `coming_soon` with a clear label.

Gating. Search, cards, the detail page, score, net yield and the full Investment Analysis are available **anonymously**. Sign-in is required to: save a property to the shortlist, save a search, send an enquiry and save a calculator scenario. Trying to save without signing in shows the toast "Sign in to save properties to your shortlist" / "Sign in to save this search".

FR-02 Unit detail

- Show price, availability, unit attributes, project facilities, approved media, score/version date, financial assumptions.
- **Developer identity is never shown.** The developer's name and contacts are not shown in the catalogue; the label is "Introduced by Best Invest", and every introduction is approved by the admin first (BR-07). The `Project.developer_visible_publicly` flag is fixed at "hidden".
- Exact unit number/address can be masked.
- An unavailable/sold/withdrawn link shows the current status and alternatives, not a generic 404. This is the `Withdrawn / 404` state with an explanation and a link to similar properties.
- The `Analysis not ready` state shows the property without the analysis block and without a score.
- The `Enquiry sent` state replaces the CTA with a confirmation and the enquiry reference.
- The CTA captures the chosen Unit.
- No listing may publish without a cover image, calculated financials and a current score.

FR-03 Analysis and score

The score has **five** categories — the same on the Investment Analysis, Compare, Score Editor and Project Review screens:

#	Category	Max points	Default weight
1	Rental Income & Net Yield	30	30%
2	Rental Demand & Tenant Quality	20	20%
3	Purchase Value & Market Position	20	20%
4	Growth & Resale Potential	15	15%
5	Risk & Investor Protection	15	15%
	Total	100	100%

- Model requirements:
 - weights total 100%; saving is blocked until the total equals 100;
 - the components add up to the displayed score (rounding tolerance ≤ 0.01);
 - every category has an explanation on the analysis screen;
 - for "Risk & Investor Protection", more points means **lower** risk — the scale direction is labelled in the UI.
- The analysis screen shows the calculation date, the model version and links to sources/disclaimer, and separates: calculated figures, deterministic explanation and the AI-assisted reviewed narrative.
- The "**Sources, assumptions & gaps**" block with provenance tags: `SOURCE` (external source), `DEVELOPER` (developer data, not independently verified), `ESTIMATE` (modelled assumption), `GAP` (missing data). The tags are mandatory — they carry legal weight.
- If AI is pending/failed, the page still renders the score breakdown (Narrative unavailable state).
- Admin confirmation required before re-scoring everything.
- Previous scores remain in history.

FR-03b Scoring rubric

Each criterion first gets a **rating from 0 to 10**, which is then converted into points by the criterion weight:

```
weighted_points = rating / 10 × criterion_weight
total_score = Σ weighted_points    (maximum 100)
```

In the MVP the weights are the same for Cyprus and Spain. What differs is the **country input data**, the legal requirements and the market benchmarks, not the weights themselves.

1. Rental Income & Net Yield — weight 30

The rating is set directly by net yield:

Net yield	Rating	Points out of 30
< 2%	0	0
2 – 2.99%	2	6
3 – 3.99%	4	12
4 – 4.99%	6	18
5 – 5.99%	7	21

Net yield	Rating	Points out of 30
6 – 6.99%	8	24
7 – 7.99%	9	27
≥ 8%	10	30

2. Rental Demand & Tenant Quality — weight 20

A structured assessment; the sub-criteria add up to a 0-10 rating:

Sub-criterion	Points
Market activity and number of comparable listings	0-3
Year-round demand	0-3
Diversity of potential tenants	0-2
Seasonality and vacancy risk	0-2

3. Purchase Value & Market Position — weight 20

The price per m² is compared with the median price of similar new builds in the district:

Deviation from the median	Rating
≥ 15% cheaper	10
10-15% cheaper	9
5-10% cheaper	8
within ±5%	7
5-10% more expensive	5
10-15% more expensive	3
> 15% more expensive	1

This needs a separate input — the **median price per m² for the district** — stored as a benchmark with its source and check date.

4. Growth & Resale Potential — weight 15

Sub-criterion	Points
Price trend	0-4
Liquidity and number of transactions	0-3
Infrastructure and economic factors	0-2
Recency and completeness of data	0-1

5. Risk & Investor Protection — weight 15

More points means **lower** risk.

Sub-criterion	Points
Legal status / title	0-3

Sub-criterion	Points
Building permits and documents	0-2
Payment / escrow protection	0-2
Developer check	0-2
Construction stage risk	0-1

Worked example

Criterion	Rating	Points
Rental Income & Net Yield (net 5.6%)	7	21/30
Rental Demand & Tenant Quality	8	16/20
Purchase Value & Market Position	8	16/20
Growth & Resale Potential	8	12/15
Risk & Investor Protection	8	12/15
Total		77/100

FR-03a Financial inputs and their approval

Source of each figure:

Figure	Source
Price, floor area, completion date	developer
Claimed rent	developer (stored separately, never used automatically)
Market rent estimate	integration with approved property portals / market-data providers
Recurring costs	approved averages for the country/region
Vacancy allowance	platform settings
Acquisition cost, gross yield, net yield	calculated automatically by formula
Cost/tax tables	official Cyprus and Spain sources, checked by a tax/legal consultant

OpenAI takes no part in producing these figures. The model only writes an explanation of an already verified result. Division of responsibility: **Make** collects data, **Bubble** stores it and calculates the formulas, **OpenAI** explains the finished result. That is why the A04 panel is called **Proposed financial estimates**, with no mention of AI.

How the market rent estimate is produced:

1. Make regularly retrieves comparable rental listings through an official API, a feed or another permitted method from approved portals for Spain and Cyprus.
2. The data is normalised by country, city/district, property type, number of bedrooms and floor area.
3. The system calculates a **recommended rental range** and a base rent estimate.
4. The admin checks the result; it is not published before this check.

For every estimate the following is stored:

- portal or data provider;
- data retrieval date;
- district;
- sample parameters;
- number of comparable listings;
- range and median;
- admin review status.

Sample acceptance thresholds. Configurable per country, because market liquidity differs.

Threshold	Default	Behaviour
Minimum listings	5	below this, the sample cannot be approved
Sufficient sample	10	between 5 and 9 the sample is approved but flagged thin sample
Maximum age	90 days	after this the sample is stale and cannot be the source of a new value

The "thin sample" flag is shown to the admin on the review screen and added for the investor as a fact tagged `ESTIMATE` — the investor should see that the estimate is based on a small sample.

Fallback. If a portal has no permitted API or feed, the MVP uses manual entry or import of comparable listings. The fallback is stored with the same set of fields and a different source value, so the provenance of the figure stays visible. The acceptance thresholds apply equally to the automatic and the manual method.

The developer's rent claim is stored separately and is never automatically used as verified market rent.

Cost/tax tables do **not** come from property portals. They are a separate source: official data for Cyprus and Spain, checked by a local legal or tax consultant. They are stored as country settings (A11), not hard-coded into the formulas.

Approval rules:

- every row has a provenance label: `Proposed` (proposed by the analyst), `Assumption` (platform default value), `Calculated` (deterministically derived from the others);
- derived values (acquisition cost, net rental income, yields, score) are **always** calculated by the deterministic service;
- proposed values do not reach the public listing until the admin approves them on A04. Before approval the panel is labelled "Not yet visible to investors";
- approval records the reviewer and time (Audit Event);
- alongside each figure, the **Financial inputs** control panel shows the source of every number ("Developer rent claim", "Our comparable rent", "Rent used for scoring").

FR-04 Calculator

- Inputs: purchase price, cash/mortgage, deposit, loan, interest, term, rent, occupancy, management fee.
- A strategy switch (`Long-term` / `Short-term`) and a rent scenario switch — **base** / **average** / **best case**. The chosen scenario substitutes its rent for the manual value.

- Country costs: transfer/VAT, legal/notary/registration and configured recurring costs.
- Validation (messages are shown under the field; the result block is not calculated while there are errors):

Field	Rule	Message
Purchase price	> 0	Enter a purchase price above zero.
Deposit (mortgage)	≥ 0	Deposit cannot be negative.
Deposit (mortgage)	$\leq$ purchase price	Deposit cannot exceed the purchase price.
Term (mortgage)	≥ 1 year	Term must be at least 1 year.
Interest (mortgage)	≥ 0	Interest cannot be negative.
Monthly rent	≥ 0	Monthly rent cannot be negative.

Occupancy (50–100%) and management fee (0–25%) are limited by the slider range, not by an error message.

- Live calculation is deterministic and uses the formula version.
- Clearly state why a personalised scenario may differ from the published yield.
- Saving/naming a scenario requires login.
- The calculator uses the same formulas as the Unit financial fields (database architecture §5.4) and the same approved rent estimate.

FR-05 Compare

- 2–3 Units; adding a fourth shows a limit toast.
- Add/remove and return to results.
- Comparison rows:
 1. Price / total acquisition cost
 2. Monthly / annual rent — base case
 3. Net yield — base case
 4. Monthly / annual rent — average case
 5. Net yield — average case
 6. Monthly / annual rent — best case
 7. Net yield — best case
 8. Gross yield
 9. Investment score 10–14. The five score categories as separate rows (points / maximum)
 10. Size
 11. Completion
- Best marker based on the configured direction; availability/risk use an explicit order, not a numeric guess.
- Mobile horizontal scroll with a pinned metric column.

FR-06 Registration and authentication

- Investor registration (screen 08) — fields:

Field	Type	Required	Error message
Full name	text	yes	Enter your full name.
Email	email	yes	This address is already registered. Sign in instead?
Password	password	yes	Use at least 8 characters, including one number.
Phone	tel	no	—
Country of residence	text	no	—

Plus consent to Terms/Privacy and a **separate** marketing consent.

- Sign-in with Google is not included in the MVP — email + password only.
- Email verification required before enquiry/contact sharing; browsing policy is configurable.
- Duplicate email, weak password, wrong credentials, locked/suspended/unverified states.
- Login (screen 15)** has three states: `Credentials`, `Two-factor`, `Locked`, plus the error state "We do not recognise this email and password combination."
- Admin login (screen 31)** — a separate login with its own two-step process (credentials → 2FA) and a locked state.
- Forgot password (screen 24)** — four steps: `Request` → `Sent` (a neutral message that does not reveal whether the account exists) → `Reset` → `Done`. Handling of an expired/used token is shown in text.
- Route by active role/workspace.
- Admin accounts require a stronger access policy/2FA, subject to Bubble capability and plan.

FR-07 Investor dashboard

- Profile criteria with an edit link: Budget, Target yield, Preferred countries, Strategy.
- Recommendations = the same catalogue ranking with saved criteria, unless a separate documented model is approved.
- Saved Units with changed price/availability badges; removing from the list has a confirmation step ("removing" state).
- Saved Searches (saved filters to reopen; no email alerts in the MVP).
- Enquiries with canonical history and public notes.
- Empty/loading/error states for each panel independently, plus a separate full empty dashboard state "First day — nothing yet".

FR-07a Account settings (screen 25)

- Contact fields: Full name (required), Email (required, duplicate check), Phone, Country of residence.
- Editable investment criteria: Budget, Target yield, Time horizon.
- Consent and notification management.

- The save button has three states: `Save changes` → `Saving...` → `Saved`; a repeated click during saving is ignored.
- Password change — through a confirmation modal warning that sessions on other devices will be ended.
- **Account closure** — a modal with a required reason and text explaining that records linked to a completed introduction are kept for seven years as required by law.

FR-08 Developer application

- Required fields explicitly marked: every field is labelled `Required` / `Optional`.
- Application fields (screen 19), three groups:

Company: Registered company name*, Registration number*, Country of registration*, Developer licence number*, Company website, Years active. **Contact:** Contact name*, Work email* (corporate domain only), Phone* (with country code), Password*, Confirm password*, Role. **Portfolio:** Projects completed, Projects currently selling, Typical unit price.

- Application documents are split into three levels:

Document	Level
Company registration certificate	REQUIRED — before verification
Developer licence	REQUIRED — before verification
Building permit for first project	TO CONFIRM — needed before project publication, not before account opening
Escrow or bank guarantee confirmation	TO CONFIRM — off-plan only
Two client or bank references	OPTIONAL — strengthens the application

The document matrix is configurable. The list is not hard-coded; it is defined by records with these parameters: country · document type · developer or project · required for account creation · required for project submission · required for publication · validity period.

The matrix is set separately for Cyprus and Spain; building permit and escrow are not hard-coded as required for all countries.

- Private uploads with type/size/progress/cancel/error.
- Application received confirmation and email with a reference like `APP-0142`.
- Applicant can resume and answer `more_info_required`. Screen 26 (Verification Status) has three states: `pending` ("Under review", decision expected within 2–5 working days), `more info required` (a list of what must be replaced, plus an upload button with progress) and `rejected`.
- Terms/listing agreement acceptance version recorded.

FR-09 Project editor

- Project fields (screen 11): Project name, Location, GPS coordinates, Completion date, Number of units.

- A project type switch: **single property** (villa/house) or **complex** with several units. For a single property the units block is replaced by one form.
- Facilities — a set of checkboxes: Communal pool, Gym, Gated area, Underground parking, Tennis / golf, Concierge, Lift, Landscaped gardens.
- Unit fields: Unit number, Bedrooms, Bathrooms, Area m², Price, Expected rent.
- Unit spreadsheet import is excluded from the first release unless separately budgeted; provide a disabled/coming-soon state or remove the link.
- Reorder photos, choose cover, assign a floor plan to a Unit Type. Floor plans (PDF/PNG) are kept in a separate list and assigned to a type or to the whole project.
- Unit create/edit/duplicate/remove with confirmations. Duplicating automatically increments the unit number and sets **Available**.
- Media upload has a percentage progress bar, cancel and an error state.
- Explicit actions, no ambiguous button: **Save draft**, **Discard draft**, **Submit for review**, each with a confirmation modal.
- The readiness checklist distinguishes "ready to submit" and "ready to publish". Four rows: Project details, Developer profile, Photos (N of 30), Units (N of declared). A project **can** be submitted with an incomplete set of units, with a warning in the modal: "12 units declared · 3 entered. You can submit and add the rest before publication." — so unit completeness is a condition of **publication**, not of submission.
- The submission confirmation says the project cannot be edited during review and gives a typical timeframe of 3–5 working days.
- Duplicate submit protected by idempotency; reference like **PRJ-0311**.

FR-10 Leads/introduction

- Before release the developer sees the enquiry reference, Unit/project, budget band, fit summary and stage without investor PII.
- The Leads screen (22) shows rows with: reference (**LEAD-0412**), project/unit, budget band, stage, contact, date. Until a decision is made, the contact field shows a lock and the text **"Contact details released after Best Invest approves"** — no PII. After approval, the name and email/phone appear in the same field.
- After contact release, show only the approved contact fields.
- The developer can acknowledge/respond and record the outcome.
- SLA age indicator after introduction. The developer has **five working days** for first contact (text of the "Approve & connect" modal).
- **Behaviour on decline:**
 - the investor receives a neutral message **without a reason** and three similar properties;
 - the developer sees **neither** the fact of the specific decline nor the reason — only a count of filtered-out requests;
 - the reason is stored in the audit log and shared only within the team.
- Introduction decisions in the A03 queue and their consequences:

Action	Emails	Result
Approve & connect	to the investor and the developer	contacts disclosed, status "CONNECTED"
Hold	none	status "ON HOLD", reason required
Decline	investor only (neutral text)	status "DECLINED", reason required

The Approve modal warns explicitly: "Contact details cannot be recalled once sent".

FR-11 Admin approvals

- One queue (A03) holds four types: project submission, developer application, introduction, listing change. **AI review is not a separate queue type** — it is the analysis panel on A04 (see §6.2).
- Type filter — five tabs; plus filters by request age, **overdue** (follow-up date in the past) and status.
- Every decision that rejects or requests changes has a reason; the modal does not allow confirming an empty reason and shows "A reason is required before this can be sent."
- The approval UI shows full context and stale-record detection.
- Publish and contact release require confirmation.
- All decisions append an Audit Event.

No Undo after a decision. After **Approve & connect** the contacts are already disclosed and the emails sent, so cancellation is impossible. Rules:

- confirmation is shown **before** sending, not after;
- after sending, **Reopen** (for Hold / Decline) or **Correct status** are available — they move the status forward rather than rolling the action back;
- no label claims that contacts or emails were recalled.

Hold and Query require three fields:

Field	Required	Note
Reason	yes	
Follow-up date	yes	date to return to the item
Owner	yes	in the MVP filled in automatically = current admin, not chosen each time

A filter by owner will be added later, when there is more than one admin.

FR-11a Admin dashboard (A02)

- Five counters: Registered investors, Active developers, Published properties, New enquiries, Hot leads. The last two are highlighted with the accent colour.
- A card for the next project awaiting approval with the actions Open full review / Approve / Request Changes / Reject.
- New enquiries table: Investor, Property, Budget, Stage, Age.
- "Nothing to review" state with a link to User Management.

Statuses: one canonical status, three sets of labels. Different words on A02, A03 and D10 are role labels for one status, not different stages. Full table:

Canonical status	Admin	Investor	Developer
submitted	New	Request received	not shown
screening	Qualifying	Awaiting analysis	not shown
qualified	Hot lead / Awaiting you	Awaiting admin approval	Awaiting admin approval
on_hold	On hold	Under review	On hold
declined	Declined	Not available	Filtered out by Best Invest
approved_for_intro	Connecting	Introduction being prepared	Introduction being prepared
introduced	Connected	Developer contacted	Introduced
developer_responded	In progress	Developer responded	Contacted
closed_won	Converted	Completed	Converted
closed_lost	Closed	Closed	Closed
cancelled	Cancelled	Cancelled	Cancelled

AWAITING YOU and Hot lead are admin labels for the qualified status, not separate stages. "Not shown" means the record is unavailable to that role at the privacy-rule level, not just hidden in the layout.

FR-11b Score model editor (A07)

- Five weight sliders, each 0-100%, step 1.
- The total is shown as a large figure; while it is $\neq$ 100, a warning is shown: "Weights must total 100 before saving. Currently over/under by N".
- A Reset button restores the default weights.
- Saving goes through a confirmation modal that states the number of properties to be recalculated.
- The recalculation has progress ("N of 83"), a completion state (with **Revert** to the previous version) and a failure state with a reference (SCORE-9C04) and the buttons "Retry the remainder" and "Roll back".
- The **Impact preview** panel shows how the scores of three properties would change.

Model version history. No large separate screen is needed. A small table or slide-out panel on the Score Model page is enough, with: version number · date · author · weights · reason for change · number of properties recalculated.

Revert **creates a new version** with the parameters of the previous one; it does not delete history (BR-04).

Weight rules:

- slider range — **0-100%**, no hard cap on a single criterion;
- one mandatory rule: **weights total 100%**; until they do, saving is blocked;
- a soft warning if any criterion exceeds **50%**: in that case one criterion determines the whole score and the other four become decorative. The warning **does not block** saving — it informs.

The 50% threshold is a UI constant; it can be changed without a database migration.

FR-11c User management (A09)

- Table: Name + email, Role, Registered, State, actions.
- Account states: `Active`, `Pending verification`, `Suspended`.
- Row action: **Suspend** (requires a reason).

"View as user" (signing in as a user) is not included in the MVP.

FR-12 Emails

All emails are sent **by Bubble** (`Send email` in backend workflows), with the platform's own SendGrid key and a domain with SPF/DKIM so emails do not land in spam. Make sends no emails.

Event	Recipient
Email verification / password reset	the user
Developer application received	the developer
Verification decision: more info required / approved / rejected	the developer
Project decision: changes requested / approved / published / rejected	the developer
Change request: queried / approved / rejected	the developer
Enquiry received	the investor
Introduction approved (Approve & connect)	the investor and the developer, each with the other's approved contact details
Enquiry <code>on_hold</code>	nobody
Enquiry <code>declined</code>	the investor only — neutral template, three similar properties, no reason
Price/availability of a unit changed	investors who saved the unit or have an open enquiry about it

Rules:

- every email is logged as an Integration Job with an idempotency key, so a retry never sends it twice; a failed email appears on the A10 Automation Monitor with a Retry button and never cancels the decision;
- introduction: two separate emails; if one fails, only that one is resent; the Enquiry moves to `introduced` when both are sent;
- decline: the reason never goes into any email; the developer receives no email and sees only a count of filtered-out requests;
- price/availability alerts are sent only to users with the matching consent; security and service emails are always sent.

9. Business rules

BR-01 Publication

A public listing exists only when the Project is `published`, the Unit has `publication_status = published`, the Unit availability is displayable, the developer is verified, required media/documents pass and a current score exists.

BR-02 Financial source

Every rent input has `source_type`, `source_reference`, `verified_at`, `verified_by`. A developer estimate is never silently treated as a verified Best Invest estimate.

This is implemented in two ways, both mandatory:

1. On the analysis screen (P04) and in the review (A04), every source row has a `SOURCE / DEVELOPER / ESTIMATE / GAP` tag.
2. In the Financial inputs panel on A04, the developer's claimed rent ("Developer rent claim €1,200 / month") is shown **next to** Best Invest's own estimate ("Our comparable rent €1,100 – €1,250") and the value actually used for the score ("Rent used for scoring").

BR-03 Contact privacy

Contact data is released only after investor consent + admin approval. Approval alone without consent is insufficient.

BR-04 Score changes

A new score model creates new scores; it never overwrites a historical Listing Score. Publishing a model requires admin confirmation.

BR-05 AI content

AI content cannot be public with a pending/rejected review or stale source versions.

BR-06 Account closure

Closure blocks access immediately, explains retained records and queues export/anonymisation according to the approved retention policy.

BR-07 Developer anonymity

The developer's name and contacts are never shown in the public catalogue or on the property page. The public label is "Introduced by Best Invest". Contacts are disclosed only through the contact release on the Enquiry, after investor consent and admin approval.

BR-08 Editable fields after publication

After publication the developer changes only `Unit.price` and `Unit.availability` (values `Available` / `Reserved` / `Sold`). Any change takes effect only after admin approval. All other fields are locked and change through a full content review.

10. Permissions and security

- All private data types receive privacy rules before real data.
- UI visibility is never the only permission control.
- Sensitive writes use backend workflows with `Only when` authorisation.
- Make calls narrow authenticated Workflow API endpoints, not the public Data API.
- The Bubble admin token, OpenAI key and Make key are private server-side secrets.
- Uploaded verification/legal files are private and attached to protected records.
- Option Sets contain no secrets or mutable country/legal configuration.
- URL/query parameters use public IDs only; never email, token, exact address or private note.
- Development and Live use separate databases and credentials.
- Run the Bubble security checklist and privacy rule tests before launch.

11. Non-functional requirements

Performance

- Landing LCP target < 2.5 s on modern mobile/4G for core content.
- Search filter update p95 < 1.5 s on the agreed test dataset.
- Dashboard initial useful content p95 < 2.5 s.
- Avoid N+1 searches in repeating groups; denormalised public fields on Unit.
- Images responsive/compressed; original assets retained only where needed.

Reliability

- Critical external side effects are idempotent.
- AI/email failure never corrupts core records.
- Every async job is visible to the admin with its retry state.
- The standard error banner includes retry and a correlation reference.

Accessibility

- WCAG 2.2 AA target for public and core authenticated flows.
- Keyboard navigation, visible focus, labels/errors associated with fields.
- Meaning is not conveyed by colour only; score/risk includes text.
- Touch targets at least 44×44 CSS px where practical.
- Tables get a mobile alternative/scroll affordance.

Responsive support

- Mobile-first 360 px, tablet 768 px, desktop 1280+ px.
- The interface switches layout at **one breakpoint — 760 px** (below it, the mobile layout). This is not a fluid grid: the mobile and desktop versions differ structurally, not just in spacing. Details are in `05-design-system.md §7`.
- Search filters become a drawer/bottom sheet with a selected count.
- The Top 5 rail moves above the result list on mobile.
- The developer portal and admin side navigation become a horizontal scrolling strip on mobile.

- The legal page contents become a horizontal strip on mobile.
- The gallery becomes a swipe carousel; the enquiry CTA may become sticky.
- Admin tables may scroll, but primary actions remain accessible.

Privacy/GDPR

- consent version history;
- data export and closure workflow;
- cookie banner/preferences for non-essential cookies;
- retention table and subprocessor list;
- no Live PII in Development without masking.

Auditability

- actor, action, entity, before/after redacted snapshot, time, source and correlation id for critical changes;
- audit records are append-only and admin-restricted;
- the displayed score/analysis is reconstructable by version.

12. Analytics events

No PII in event properties.

Event	Key properties
search_started	country, price band, source
search_filter_changed	filter key, result count band
unit_viewed	unit public id, position, search id
compare_added	unit public id, count
registration_started/completed	role, source
unit_saved/unsaved	unit public id
analysis_viewed	model version, narrative state
calculator_used/saved	financing mode
enquiry_started/submitted	unit public id, source
developer_application_submitted	country count, documents complete
project_submitted	units count band, media count band
approval_decided	type, decision, age band

The analytics provider and consent mode are defined before implementation.

13. Delivery plan

Phase 0 — Data preparation

Outputs:

- cost/tax tables for Cyprus and Spain;
- document matrix per country;
- list of portals for collecting comparable rental listings;
- legal texts, retention and consent.

Exit: frozen v1 schema and seed data.

Phase 1 — Foundation

- Bubble data types, option sets, privacy rules;
- authentication/roles;
- Country Config, costs, score model;
- design system and standard state components;
- audit/integration job framework;
- Dev/Live credentials.

Phase 2 — Investor core

- P01-P06;
- deterministic financial/score services;
- responsive catalogue and public detail;
- registration/login/reset.

Phase 3 — Portals and approvals

- investor dashboard/settings/saves/enquiries;
- developer application/status/portal/project editor;
- admin verification/project review/approvals;
- end-to-end introduction.

Phase 4 — Make/OpenAI/notifications

- integration scenarios;
- AI review queue and fallback;
- transactional email;
- alerts and operational monitor.

Phase 5 — QA, legal, launch

- privacy/security tests;
- browser/mobile/accessibility QA;
- data seed and migration rehearsal;
- AI eval;
- load/workload test;
- legal content sign-off;
- incident/rollback runbook;
- production smoke test.

14. Test strategy

Functional

- happy path and all status branches for investor/developer/admin;
- required/optional validations;
- all empty/loading/error/permission states;
- back/refresh/deep-link/session-expiry behaviour;
- duplicate submit and stale admin decision.

Security

- anonymous cannot query private fields/files;
- investor A cannot see investor B data;
- developer company A cannot see company B projects/docs/leads;
- a masked lead cannot reveal PII through search, URL, API response or page source;
- role escalation via UI/API is blocked;
- a Make callback with the wrong job/entity/key is rejected;
- public file URLs tested both authorised and unauthorised.

Data integrity

- Unit cached fields match Project/Unit Type after rebuild;
- published filters return only allowed records;
- financial formula golden tests;
- score component/weight totals;
- a Change Request applies once;
- historical versions stay unchanged.

Integration

- retry, timeout, 429, partial email, callback failure, stale AI, duplicate webhook;
- Dev cannot modify Live;
- secrets absent from client code/log output.

15. Definition of Done

A feature is complete only when:

- the functional requirement and role permissions are implemented;
- loading, empty, error, success and unauthorised states are present;
- server-side validation and the privacy rule are verified;
- audit/integration events are added where required;
- responsive mobile/desktop is checked;
- accessibility basics are checked;
- the analytics event is documented/implemented if applicable;
- the acceptance test passes in Development and a staging/live-safe environment;
- the related specification is updated.

16. Launch acceptance criteria

1. An investor completes search → unit → registration → enquiry without manual database intervention.
2. A developer completes application → approval → project submission → changes → publication.
3. The admin can inspect full context before every approval.
4. Contact data remains masked until explicit release and is delivered once.
5. Published figures are traceable to input/version and match the calculator rules.
6. AI failure degrades to deterministic analysis, not a broken page.
7. All critical Make scenarios recover from transient errors without duplicates.
8. Private documents fail access tests for anonymous and unrelated roles.
9. Account settings cover preferences, an export request and a closure request.
10. Legal pages, cookies and disclaimers have a named owner and approval date.
11. No prototype-only/superseded/internal pages are public.
12. The admin automation monitor shows failed/stuck jobs with correlation IDs.

17. Out of scope for the MVP

- multiple team members per developer company / invitations (one user per company in the MVP);
- in-platform chat;
- CRM sync/export;
- map search and location intelligence;
- investor conversational AI;
- mortgage/lender referral;
- multi-currency and full localisation;
- transaction/success-fee/invoice accounting;
- public partner API/feed;
- investor post-purchase portfolio tracking;
- bulk spreadsheet unit import unless separately estimated;
- advanced analytics dashboard;
- saved-search email digest.

18. Risks

Risk	Impact	Mitigation
Unverified tax/cost assumptions	misleading net yield	versioned country rules + expert approval
Bubble privacy misconfiguration	PII/document exposure	privacy-first schema + role penetration tests
Duplicate execution (Make or Bubble)	duplicate emails/contact release	Integration Job + idempotency key
AI unsupported claim	reputational/legal	structured facts, validation, human review, fallback
Different status labels per role	broken journey/reporting	one canonical Enquiry Status
Published edits without history	audit failure	Change Request + Audit Event

Risk	Impact	Mitigation
Heavy live searches/workload	cost/performance	denormalised Unit fields, constraints, pagination
Legal text remains draft	launch blocker	named counsel/owner and approval gate

19. Official implementation references

- Bubble: [Privacy rules](#), [API workflows](#), [API authentication](#), [API Connector security](#), [private files](#), [option sets](#), [search optimization](#).
- Make: [Webhooks](#), [Scenario settings](#), [Incomplete executions](#), [Retry error handler](#), [Data stores](#).
- OpenAI: [Models](#), [Responses API](#), [Moderations](#).